

ATHARNA

Smart Museum Security & Visitor Experience System

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Program Track: SIC7 IoT



ATHARNA

Real-World Motivation

- In September 2025, a rare **golden bracelet** was stolen from the Egyptian Museum in Cairo by a trusted expert.
- This event showed major weaknesses in how the museum watched things, managed who could access what, and kept its artifacts safe.
- It highlighted the urgent need for automatic, instant monitoring to protect valuable items from both inside and outside dangers, moving past old-fashioned manual systems.



Key Problems for Museums

Today's museums face big problems with keeping things safe, running smoothly, and making visits fun for people.

Boring Visitor Experience

- Not enough fun, digital ways for visitors to interact.
- This means visitors just look without really learning, and it's hard for them to get the full story behind the exhibits.

Weak Security

- No smart, instant ways to track valuable items and stop them from being moved without permission.
- Too much trust in old security systems that are easy to trick.

Poor Operations

- Everything from visitor entry to ticket booking and crowd tracking is done by hand.
- This causes long lines, bad record-keeping, and no clear idea of how many people are in the museum at any time.

Environmental Dangers

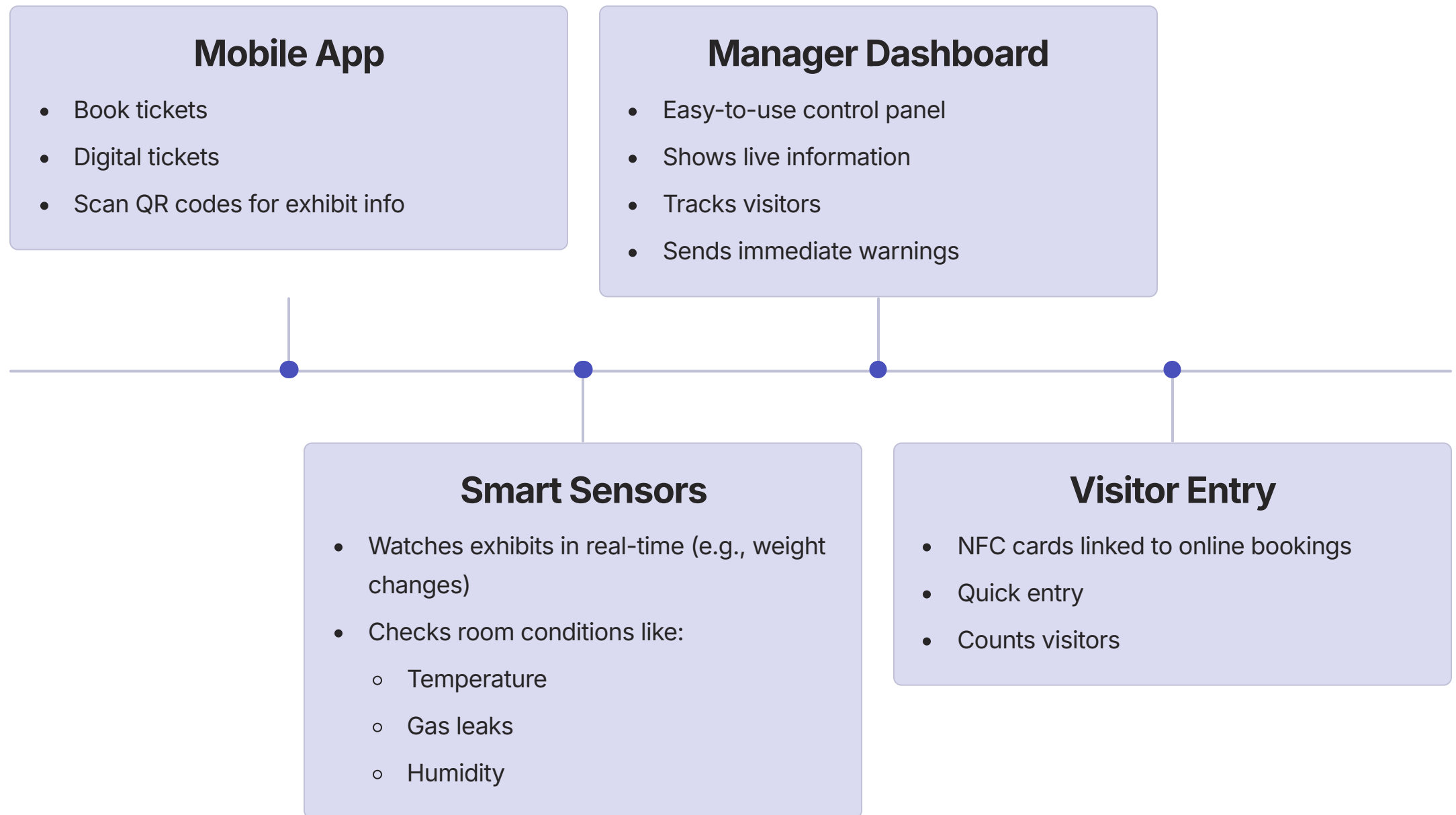
- Slow or missing warnings for serious risks like fire, gas leaks, or wrong temperature/humidity.
- Failing to quickly fix these problems can harm the exhibits.

Our Goal:

- Create a smart, automatic, and very secure museum system.
- Protect artifacts from dangers both inside and outside the museum.
- Make the visitor experience much better and more modern.

ATHARNA: A Smart System for Museums

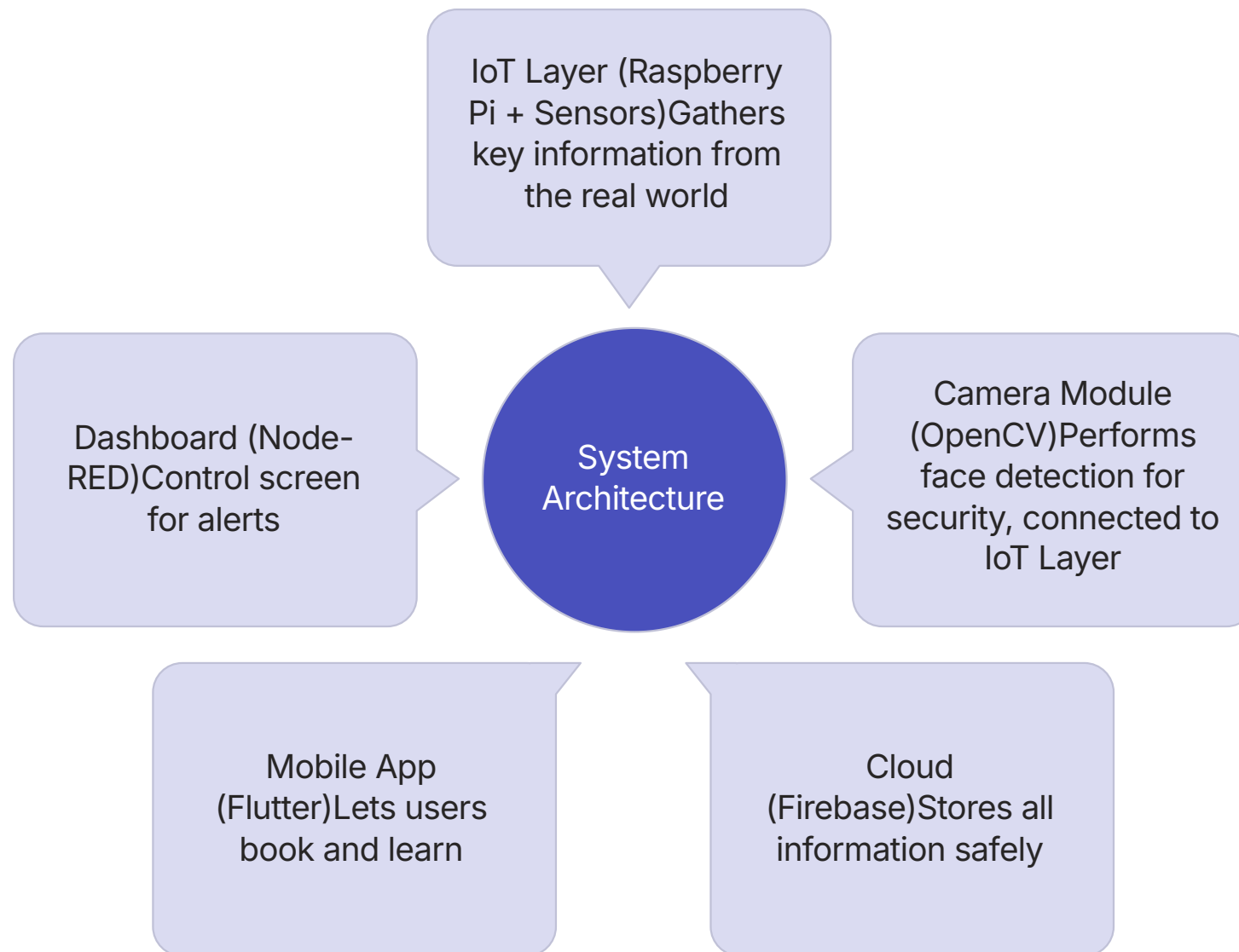
- ATHARNA solves problems by connecting devices, cloud services, and mobile apps.
- It offers one place to manage security and improve the visitor experience.



- This system makes the museum a **safe, engaging, and smart** place, driven by information.

System Architecture and Data Flow

The system uses a simple, three-part structure that sends information reliably and allows for instant control. It now includes a camera module for advanced security features.



Hardware and Software Components

Hardware: The Physical Parts



Raspberry Pi

- Controls everything
- Uses Python to gather sensor data



Load Cell / Ultrasonic Sensors

- Spots if artifacts are moved without permission
- Notices if items are taken or change weight



DHT11 & Gas Sensor

- Checks temperature and humidity
- Detects smoke and gas leaks



NFC Module & Buttons

- Automatically tracks visitors
- Controls who enters and leaves



Camera Module (USB or Pi Camera)

Captures real-time video for face detection.

Software: The Smart Brain



IoT Layer (Python)

Gathers all sensor data, starts voice narrations, and sends updates to Firebase.



Mobile App (Flutter)

Offers an easy-to-use experience for visitors, handles login, bookings, and QR scanning.



Dashboard (Node-RED)

Shows live data (like temperature, visitors, and status) and sends sound/visual alerts to staff.



Database (Firebase)

A safe and flexible place to store visitor info, booking details, and sensor records.



OpenCV (Python Library)

Performs face detection and security verification that integrates with Node-RED and Firebase.

How Visitors Use ATHARNA

ATHARNA offers an easy, digital experience for visitors while keeping everything very secure.

- 1

1. Book & Get Code

 - You sign up and book your visit using our mobile app.
 - You get a special QR code for your booking.
- 2

2. Enter & Get Card

 - Your QR code is checked at the entrance, and your face is verified for authorized entry.
 - You receive a special card (NFC card) to get physical access.
- 3

3. Live Monitoring

 - The main screen (dashboard) updates with visitor count and security status.
 - Cameras continuously monitor visitors and verify authorized personnel, while sensors watch artifacts.
- 4

4. Explore & Learn

 - You scan QR codes on artifacts.
 - You get detailed history and spoken guides.
- 5

5. Security Alert

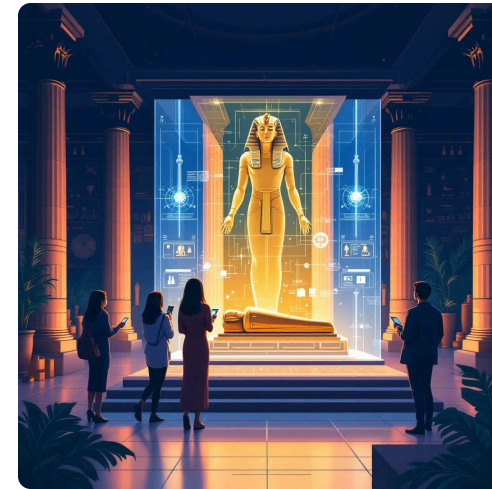
 - If an artifact is moved or taken (weight changes to zero), or if an unauthorized person is detected.
 - The system immediately sends an alert to the main screen (dashboard).

Live Numbers on the Dashboard:

25.5°C	55%	127	Safe
Temperature	Humidity Level	Visitors Inside	Artifact Status
Checks how well the AC is working.	Very important for keeping old papyrus safe.	How many visitors are inside right now (counted by card scans).	Status regarding artifact movement (Safe, Alert, or Emergency).

Conclusion

- ATHARNA successfully combines keeping things safe with digital visitor interaction, creating a strong base for running a modern museum.
- ATHARNA connects **history and technology**.
- By keeping valuable items safe and making the visitor experience modern, the system directly helps with **Egypt's 2030 vision** to bring digital ways into culture and tourism.
- The camera system has been thoroughly tested and successfully detects both authorized and unauthorized faces, providing real-time alerts directly on the dashboard.



Future Work

- Upgrade to advanced face recognition capabilities using AI models for enhanced security and visitor flow management.
- Implement cloud database storage for scalable, secure, and accessible data management.

Business Model Part

Design Thinking Process:

We used a careful design thinking process to build ATHARNA, making sure the final product truly helps museums and visitors.



1. Understand Users

- Found their main problems and saw where things weren't working well.

2. Define Problems

- Clearly described the main problems: keeping artifacts safe, making visitor movement smoother, and offering more digital learning tools.

3. Brainstorm Ideas

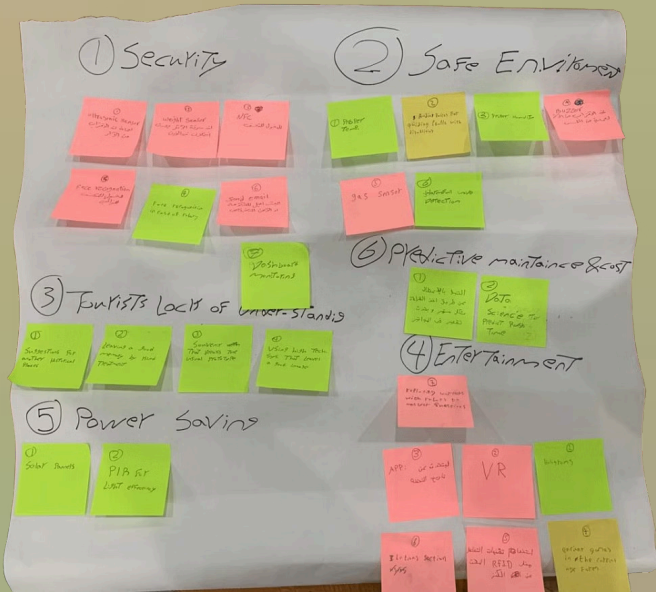
- Thought of many tech ideas.
- Decided on a single system combining smart sensors (IoT), mobile apps, and cloud technology.

4. Create Prototypes

- Built a working model using Raspberry Pi.
- Used Flutter to make the mobile app.
- Used Firebase to store data online.

5. Test Solutions

- Checked if the system worked well and responded quickly.
- Tested it in museum-like settings to make sure it was dependable and correct.



Value Proposition Canvas

ATHARNA delivers a powerful value proposition by aligning our solutions directly with the needs of our diverse customer segments.

Customer Profile

Customer Jobs

- Protect artifacts
- Manage visitors efficiently
- Offer engaging experiences

Customer Pains

- High theft risk & security concerns
- Outdated management systems
- Manual booking processes
- Lack of real-time data

Customer Gains

- Smart automation & efficiency
- Live data for informed decisions
- Enhanced visitor experience
- Improved artifact preservation

Value Map



Pain Relievers

- Sensors detect movement, weight loss, and environmental hazards instantly.
- NFC cards and dashboards track visitor count automatically for crowd control.
- Automated alerts notify staff of security breaches in real-time.



Gain Creators

- QR-based information access and personalized voice narration enrich learning.
- Easy app-based booking and digital tickets streamline entry.
- Real-time environmental data helps preserve delicate artifacts.

Our core value: ATHARNA makes museums **secure, smart, and interactive**, preserving Egypt’s cultural heritage through modern technology.

ATHARNA's Business Model

ATHARNA's business plan is built to:

- Last a long time and make a real difference.
- Connect important partners, tasks, and what we offer.

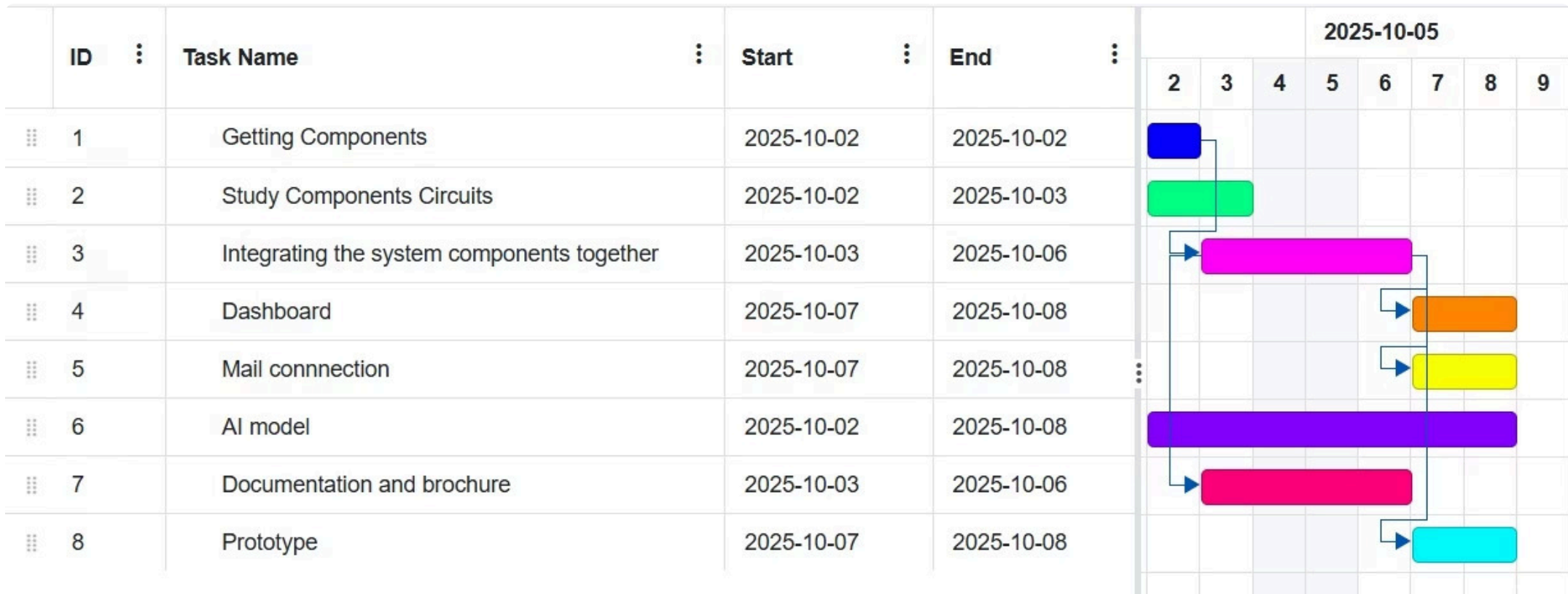
Key Partners	Key Activities	Value Proposition
• Tourism & Antiquities Ministry • Museums • Tech companies	• Setting up the system • Keeping it running • Analyzing information	• A smart, safe, and modern museum visit
Customer Relationships	Customer Segments	Channels
• Support • App messages • Feedback system	• Museums • Visitors • Government groups	• Mobile app • Tourism websites • Museum partnerships
Key Resources	Cost Structure	Revenue Streams
• Smart sensors (IoT) • Online computing (Cloud) • Development team	• Equipment • Online computing (Cloud) • Maintenance • Marketing	• Membership fees • Government money • Selling data insights

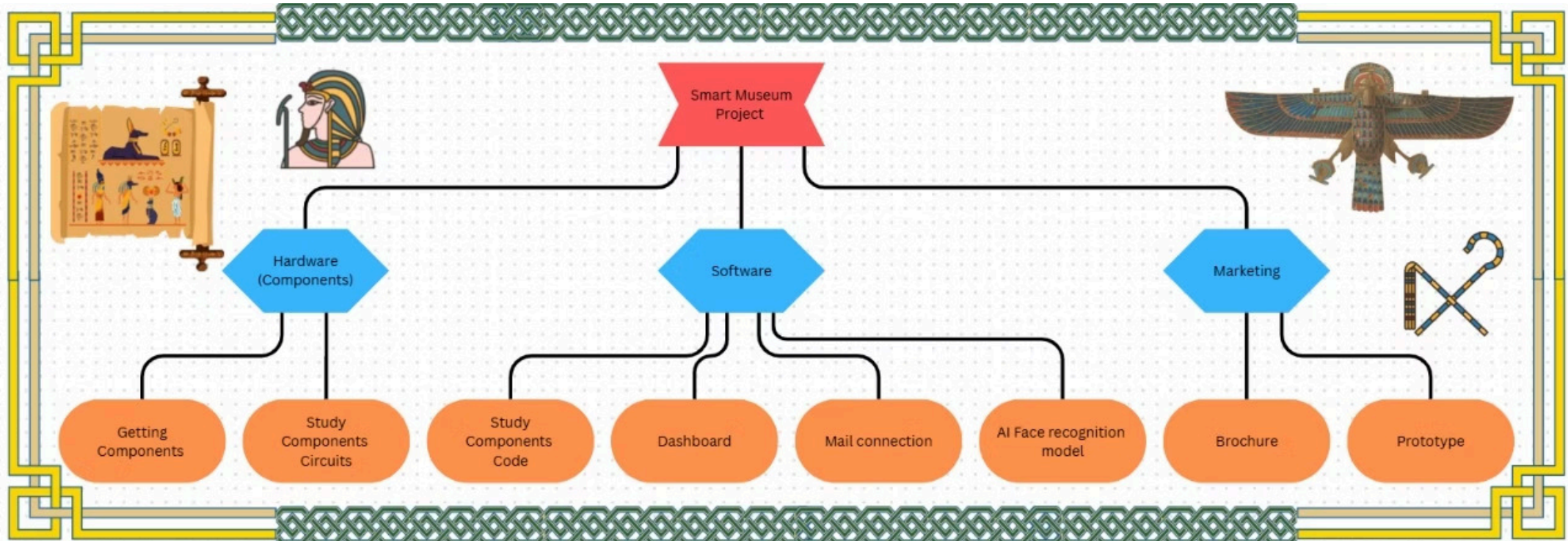
This plan will:

- Ensure a strong way to give value to everyone involved.
- Bring in steady money for future growth and improvements.

Project Management Part







Marketing Part



SIC Graduation Project
Team 2 (702)

- Three-level security with a lockdown feature
- IoT integration in the museum
- Dashboard for monitoring and control
- Online booking to skip queues



Book your ticket Now!!!

This Project Offers

- Three Security Levels
- Temperature Control
- Humidity Control
- Gas Control
- Visitor Count Control

ATHARVA

When history meets future



